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NPS/002/002 – Technical Specification for Industrial and Insulating Tapes and Mastic Compounds

1. Purpose

The purpose of this document is to detail the requirements of CE Electric UK in relation to the products specified within this document.

This document supersedes the following documents, all copies of which should be destroyed.

Ref.	Version	Title
NPS/002/002	2	Specification for Industrial and Insulating Tapes

2. Scope

This document applies to the specification required and describes in detail the requirements of CE Electric UK with respect to: -

- Anti-Corrosion Tape
- Insulating Tape
- Identification Tape
- Self Amalgamating Tape
- Electrical Insulating Putty/Tape
- Moisture Sealing Mastic Compound
- Adhesive Tape
- Vinyl Mastic Tape
- Glass Fibre Tape
- Butyl Mastic Tape
- Cotton Tape
- Solid Setting Putty

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3. Technical Requirements

3.1 Anti-Corrosion Tape

The Fabric Grease Impregnated Anti-Corrosion Tape shall be used as corrosion protection on electrical plant and associated equipment. The tape shall provide corrosion resistance in all weathers and shall be used above ground exposed to the atmosphere or directly into various ground conditions. The tape shall be resistant to UV rays, abrasion, corrosion, alkalis & acids.

3.2 PVC Insulating Tape.

For outdoor, general use as primary insulation on electrical terminations, cables etc. up to 600V. The tapes shall conform in all weather applications. The tape shall also be colour fade resistant, resistant to UV rays, abrasion, corrosion, alkalis & acids. The tape shall adhere to PVC and metal surfaces and have pliability and adhesion at 0°C

3.3 Coloured Identification Tape

For identification purposes on electrical terminations etc. the tapes shall be colour fade resistant, resistant to UV rays, abrasion, corrosion, alkalis & acids. The tape shall adhere to PVC and metal surfaces and have pliability and adhesion at 0°C.

3.4 Self-Amalgamating Tape

The Self amalgamating tape shall be based on Polyisobutylene (PIB). The tape shall be interleaved with an easy to remove plastic film and must be non-tacky, easy to handle and apply. The tape shall amalgamate rapidly when applied under tension and provide a void free homogenous wrapping without the need for external heat or pressure. It shall be compatible with polyethylene, cross-linked polyethylene, ethylene propylene rubber, PVC, butyl and neoprene cable insulation and polyurethane resins. The tape must be highly resistant to prolonged immersion in water, resistant to ozone and compatible with hot pouring compounds up to 125°C used in cable joint-boxes.

3.5 Electrical Insulating Putty / Tape

Rubber insulating compound in tape form, backed by a removable fabric. The tape shall be permanently elastic, non-hardening and highly resistant to chemical corrosion. The tape will be used for filling voids and irregular shapes in some terminations and joints (not oil filled or oil impregnated cables).

3.6 Moisture Sealing Mastic Compound

The compound shall be self supporting, moisture sealing media, mainly for the sealing of cable entry ducts to buildings. The compound shall be saturated petroleum hydrocarbons, fibrous filler and thermal extender, which shall be permanently plastic, non-cracking and resistance to moisture, mineral acids and salts. The compound shall not oxidise and shall contain no evaporating solvents.

3.7 Black Adhesive Tape

The tape shall be used to provide temporary protection to PVC oversheaths during sheath removal or to build up cable diameters to suit joint shells etc. The tape shall be of a cotton cloth material with a blended rubber adhesive insulating compound coated on both sides.

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3.8 Glass Fibre Adhesive Tape

The tape shall be white with thermosetting pressure sensitive adhesive and designed to perform continually at 130°C. The tape shall have good mechanical strength, be waterproof, shrink, rot and stretch resistant in the most arduous conditions. The tape must adhere to any clean dry surface and have good adhesion over a range of ambient temperatures between 0°C and 30°C.

3.9 Miscellaneous Items

- a) Tape Vinyl Mastic: Self-fusing rubber based insulating compound laminated to a flexible, all weather grade vinyl (PVC) backing.
Main usage: - To insulate and moisture-seal all connections up to 600volts.
- b) Butyl mastic tape: Non curing type that shall retain the same properties and consistency after ageing in different environments.
- c) Semi-conducting: Ethylene Polpylene Rubber Tape.
Usage: - too electrically round out high voltage connectors and to minimise electrical stresses (Scotch 13 Type or equivalent).
- d) Aluminium Oxide Tape: 25mm x 50M: Grade P- 400.
- e) Identification Tape: Black "N" on Yellow Background: Self - Adhesive: CLOTH: 17mm Wide.
- f) Solid Setting Putty 2-Part Mixture (Resin & Hardener): Used for sealing, repairing, insulating and waterproofing a variety of materials including metal, wood glass and some plastics.

The tapes shall conform in all weather applications. The tapes shall also be colour fade resistant, resistant to UV rays, abrasion, corrosion, alkalis & acids. They should be suitable for outdoor and general use.

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4 References

4.1 External Documentation

The tapes shall comply with the relevant International Standards, British Standard Specifications and all relevant Energy Network Association Technical Specifications (ENATS) current at the time of tendering, except where varied by this standard. In respect the following documents are particularly relevant

Reference	Title
BS 3924	Pressure-Sensitive Adhesive Tapes for Electrical Insulating Purposes
BS EN 1939	Self-adhesive tapes. Determination of peel adhesion properties
BS60454 -1	Pressure-sensitive adhesive tapes for electrical purposes. General requirements

The supplier shall provide with the tender full technical details of the equipment offered and shall indicate any divergence from these standards or specifications.

4.2 Amendments from Previous Version

Clause	Subject	Amendments
2.1	Contents	Inclusion of contents page into document
4.2	Amendments	Inclusion of amendments list into document
Appendix 1	Requirements	Removal of duplicate Item
3.9 (f)	Requirements	Inclusion of additional item
Appendix 1	Requirements	Add clause number to item description/commodity code

5 Definitions

Term	Definition
NIL	

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6 Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
R McMahon	Standards Engineer	R McMahon	22/02/2010

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
M Nicholson	System Strategy	M Nicholson	18/03/2010

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
V Gorny	Standards Engineer	V Gorny	03/03/2010

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S. Johnson	CDS Administrator	S. Johnson	29/03/2010

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C. Holdsworth	Standards and Assessment Manager	C. Holdsworth	11/03/2010

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C. Morgan	Head of Procurement	C. Morgan	16/03/2010

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Appendix 1 – Requirements

Clause	DESCRIPTION	Catalogue Number
3.1	Fabric Grease Impregnated Anti-Corrosion Tape: 50mm x 10m (Denso)	062661
3.2	Insulating Tape: PVC Self Adhesive Colour BLACK : 19mm Wide.	063529
3.3	Identification Tape: PVC Self Adhesive Colour RED : 19mm Wide.	063412
3.3	Identification Tape: PVC Self Adhesive Colour BLUE : 19mm wide.	063446
3.3	Identification Tape: PVC Self Adhesive Colour YELLOW : 19mm wide.	063438
3.3	Identification Tape: PVC Self Adhesive Colour GREEN : 19mm wide.	063453
3.3	Identification Tape: PVC Self Adhesive Colour BROWN : 19mm wide.	350960
3.3	Identification Tape: PVC Self Adhesive Colour GREY : 19mm wide.	350961
3.4	Self Amalgamating Tape: Colour BLACK : 50mm Wide.	064220
3.5	Rubber insulating compound in tape form	353871
3.6	Mastic Compound: 330mm x 50mm	089896
3.6	Mastic Compound: 3kg blocks	353208
3.7	Tape Black Cotton Cloth Adhesive Coated 19mm X 20M	350498

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Clause	DESCRIPTION	Catalogue Number
3.8	Glass Fibre Tape Adhesive: 25mm x 50M (AG Tape)	062752
3.8	Glass Fibre Tape Adhesive: 100mm x 50M (AG Tape)	062778
3.9 (a)	Tape Vinyl Mastic: 38mm x 2.3mm x 6M	064832
3.9 (a)	Tape Vinyl Mastic: 102mm x 2.3mm x 3M	064840
3.9 (b)	Butyl Mastic Tape: 25mm x 5M (Black)	064857
3.9 (d)	Aluminium Oxide Tape: P-400	350959
3.9 (c)	Semi-Conducting: 20mm x 5M (Scotch fill 13 Type) (For Use On HV Overhead Conductors)	365328
3.9 (e)	Identification Tape: Black "N" on Yellow Background: Self - Adhesive: CLOTH: 17mm Wide.	350963
3.9 (f)	Solid Setting Putty 2-Part Mixture (Resin & Hardener)	351931

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Appendix 2 – Logistical Requirements

To enable CE Electric UK to store the product(s) in accordance with the manufacturer's recommendations the Tenderer should provide details of the recommended storage environment with respect to each tendered product.

Details should be provided where relevant in respect to the minimum and maximum exposure levels, frequency of exposure and duration of exposure of the packaged item with respect to;

- | | | | |
|---|---------------------|---|-----------------------|
| * | Ambient temperature | * | Atmospheric corrosion |
| * | Humidity | * | Impact |
| * | Water | * | Vibration |
| * | Dust | * | Solar radiation |

The Tenderer shall ensure that each item is suitably packaged and protection to maintain the product and packaging as "fit for service" prior to installation taking account of the potential for an outdoor storage environment. All packaging shall be sufficiently durable giving regard to the function, reasonable use and contents of the packaging. Tenderer shall submit at the time of tendering the details of the proposed packaging (i.e. materials composition and structure) to be used for each product.

Palletised goods shall be supplied on standard 1200mm x 1000mm pallets.

Clearly legible, easily identifiable, durable and unambiguous labelling shall be applied to each individual and where relevant multiple package of like products. As a minimum requirement the following shall be included:

- * Manufacturer's trademark or name
- * Supplier's trademark or name
- * Description of item
- * Date of packaging and/or batch number
- * CE Electric UK product code
- * Weight
- * Shelf life of the product

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Appendix 3 - SELF CERTIFICATION CONFORMANCE DECLARATION

Specification for Industrial and Insulating Tapes required to be supplied against this specification shall comply with the latest issues of the relevant ENATS, British and International Standards specified. The following tables are intended to amplify and/or clarify the requirements of elements of these Standards but do not preclude meeting all requirements of the standards. .

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Instructions for completion

- When Cs1 code is entered no remark is necessary.
- When any other code is entered the reason for non-conformance shall be entered.
- Prefix each remark with the relevant 'BS EN' 'IEC' or 'ENATS' as appropriate.

Manufacturer:

Product Reference:

Details of the Cable Type (Voltage, Conductor Type and Size)

Name:

Signature:

Date:

NOTE:.

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Specification for Industrial and Insulating Tapes				
		Clause / Requirements	Conformance Code	Remarks / Comments
Anti-Corrosion Tape				
General		NPS/002/002 3.1		
Carrier		Fabric of high strength and chemical resistance		
Impregnant		Petroleum Jelly based corrosion inhibitor		
Weight		1.4Kg/sq M Average.		
Breaking Strength		50 N/cm width minimum		
Electrical Breakdown		5kV/mm minimum		
Insulation Resistance		5 x 10 ⁶ Megohms/25mm		
PVC Insulating Tape				
General		NPS/002/002 3.2		
Classification		3 (plastics) group 31/90 plasticised PVC		
Breaking Strength		35 MPa		
Elongation		150%		
Electrical Breakdown		40kV/mm		
Adhesion - Peel		2.8 (steel)		
		1.6 (backing)		
		Flagging - 2mm maximum		
PH of Water Extract		5.5 - 8		
Water Conductivity Extract		2.5mS/m maximum		
Freedom from Corrosive Sulphur		As in specification		
Insulation Resistance		100 000 Megohms/25mm		
Electrolytic Corrosion		CLF to be mutually agreed		
Resistance to Flame Propagation		Category 1 or 2		

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Specification for Industrial and Insulating Tapes				
		Clause / Requirements	Conformance Code	Remarks / Comments
	Coloured Identification Tape			
	General	NPS/002/002 3.3		
	Classification	3 (plastics) group 31/90 plasticised PVC or 30/90		
	Breaking Strength	30 MPa		
	Elongation	100%		
	Electrical Breakdown	40kV/mm		
	Adhesion - Peel	2.0 (steel)		
		1.5 (backing)		
	PH of Water Extract	5.5 - 8		
	Freedom from Corrosive Sulphur	As in specification		
	Insulation Resistance	100 000 Megohms/25mm		
	Self-Amalgamating Tape			
	General	NPS/002/002 3.4		
	Breaking Strength	50 N/cm width minimum		
	Electrical Breakdown	20kV/mm		
	Electrical Insulating Putty/Tape			
	General	NPS/002/002 3.5		
	Temperature Range	-10 ⁰ c to 80 ⁰ c		
	Electrical Breakdown	5kV/mm		
	Insulation Resistance	5 x 10 ⁶ Megohms/25mm		

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Specification for Industrial and Insulating Tapes				
		Clause / Requirements	Conformance Code	Remarks / Comments
	Moisture Sealing Mastic Compound			
	General	NPS/002/002 3.6		
	Composition	Saturated Petroleum Hydrocarbons, Fibrous Filler and Thermal Extender		
	Characteristics	Permanently plastic, Non-cracking and resistance to moisture, mineral acids and salt. Will not contain Oxidise and contains no evaporating solvent.		
	Storage Life	Unlimited at normal temperatures		
	Temperature Range	-10 ⁰ c - 70 ⁰ c		
	Black Adhesive Tape			
	General	NPS/002/002 3.7		
	Breaking Strength	80N/cm (20kg per 25mm)		
	Glass Fibre Adhesive Tape			
	General	NPS/002/002 3.8		
	Adhesion Temp Range	0 ⁰ c to 30 ⁰ c		
	Max Continuous Operating Temperature	130 ⁰ c		
	Characteristics	Good Mechanical Strength, Waterproof, Shrink, Rot and Stretch Resistant.		

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Specification for Industrial and Insulating Tapes				
		Clause / Requirements	Conformance Code	Remarks / Comments
	Miscellaneous Items			
	General	NPS/002/002 3.9		
	Vinyl Mastic Tape	Self-fusing rubber based insulating compound laminated to a flexible, all weather grade vinyl (PVC) backing		
	Butyl mastic tape	Non curing type that shall retain the same properties and consistency after ageing in different environments		
	Semi-conducting	Ethylene Polypylene Rubber Tape (Scotch 13 Type)		
	Aluminium Oxide Tape	25mm x 50M: Grade P -400		
	Solid Setting 2-part Putty	Temperature withstand -20 ⁰ C - +80 ⁰ C Suitable for use on wood, metal, some plastics and Glass.		